ITALIAN FINANCIAL CHALLENGE

AI-Powered Financial Health Classification

LUISS Guido Carli - Grand Challenge
March 23, 2026
Prof.ssa Finocchi Irene and Prof. Poccianti Cosimo

Direnzo Carla *Moriondo Tommaso* *Pazienza Lorenzo* *Raimondo Eleonora*

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AGENDA

1. Challenge Definition
Objective, business context, success metrics

2. Technical Setup & Infrastructure
Panel structure, temporal constraints, anti-leakage controls

3. Exploratory Data Analysis
Class imbalance, survivorship bias, structural patterns

4. Key Findings & Modeling Strategy
Critical insights, feature engineering, final approach

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CHALLENGE OVERVIEW

Objective

Classify Italian Companies into
4 financial health categories (A→D)

A = Excellent
B = Good
C = Moderate Risk
D = High Risk/Distressed

Business Context

Mirrors real-world credit rating systems
used by banks and investors

- Misclassifying D as A/B → loan defaults, portfolio losses
- Recall on Class D is critical business priority

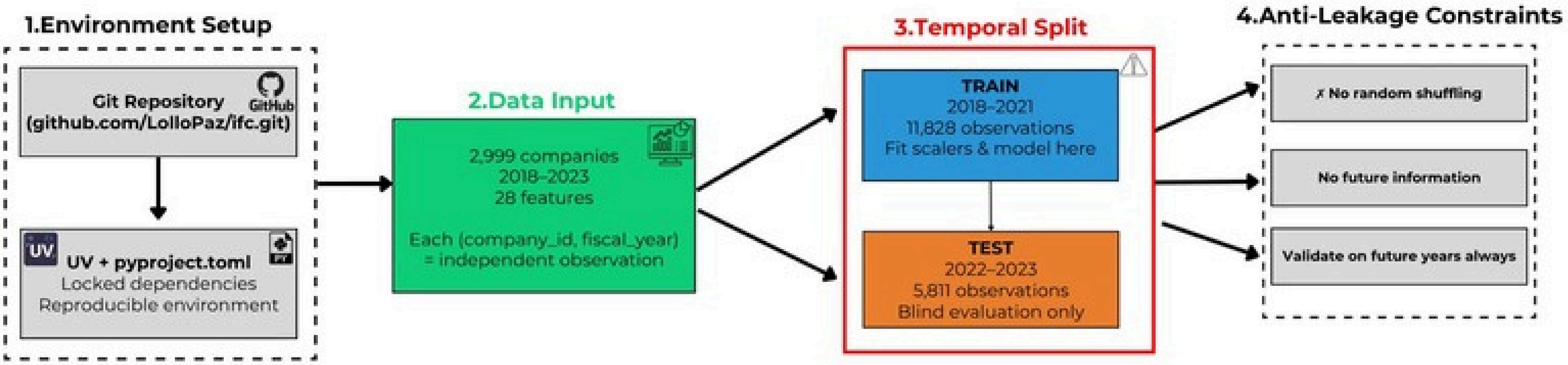
Training Instance Definition

Each row = independent diagnostic
snapshot (company, year) → (X, y)

Success Metrics

Primary: Weighted **F1-Score ≥ 0.65**
Avoid severe misclassifications (A↔D errors)
Interpretable features for business stakeholders

PROJECT SETUP & TECHNICAL INFRASTRUCTURE



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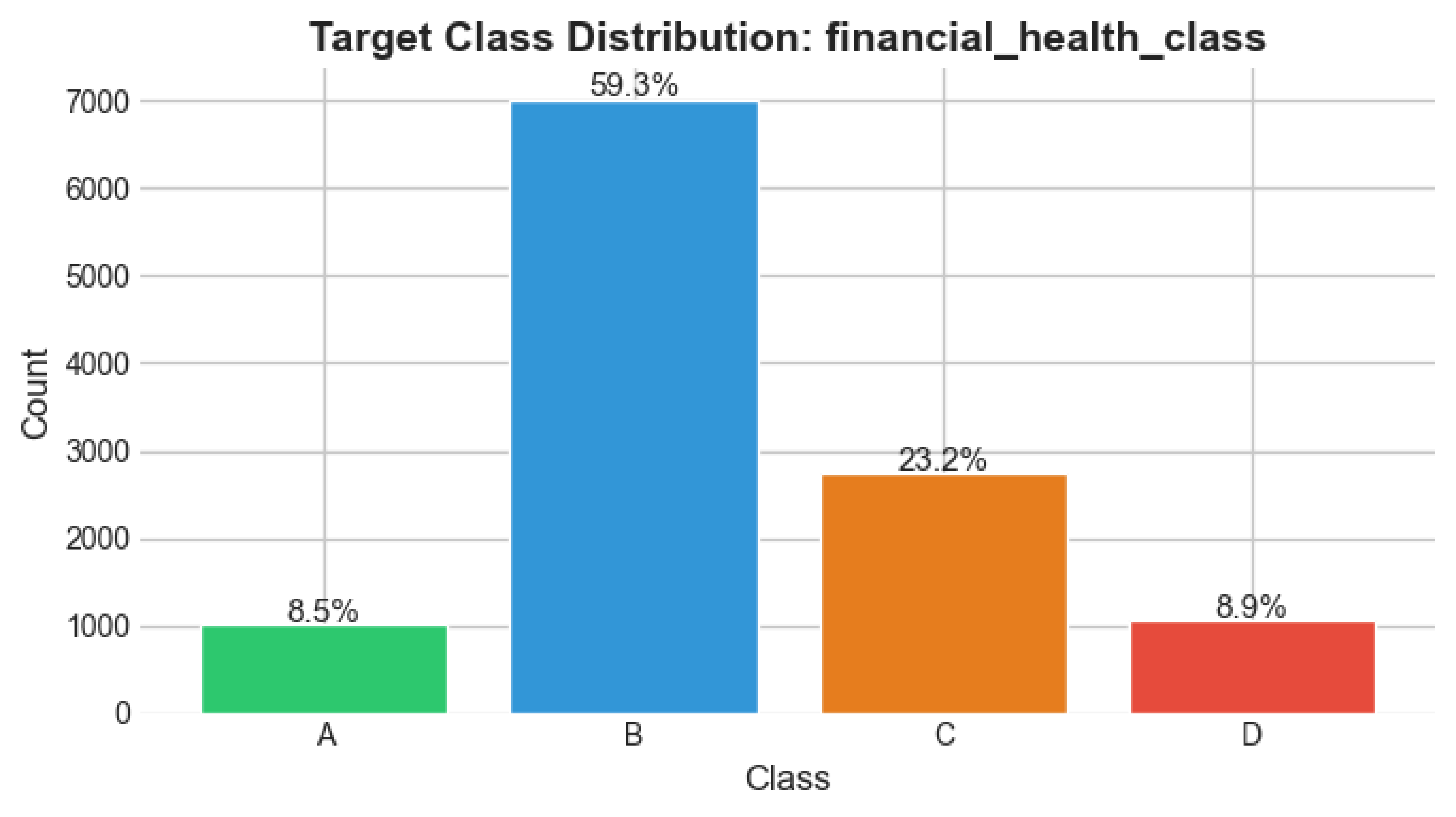
CLASS DISTRIBUTION & IMBALANCE



MODELING IMPLICATION

- Class B dominates at 59.3%
- Weighted F1-Score as primary metric
- class_weight="Balanced" strategy

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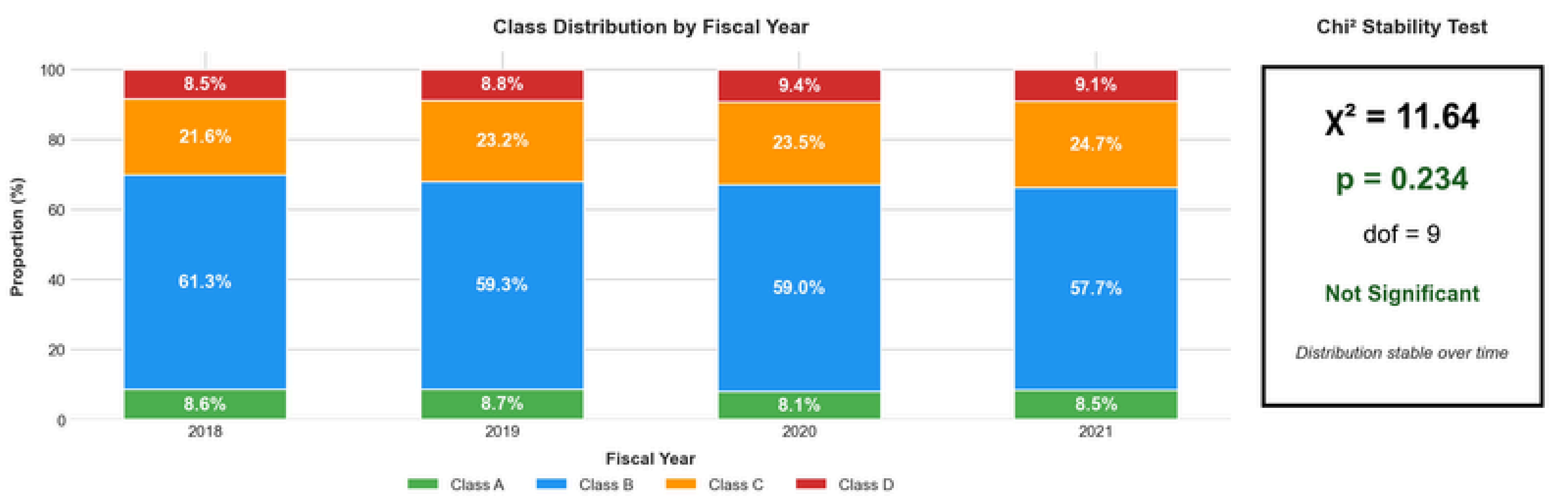
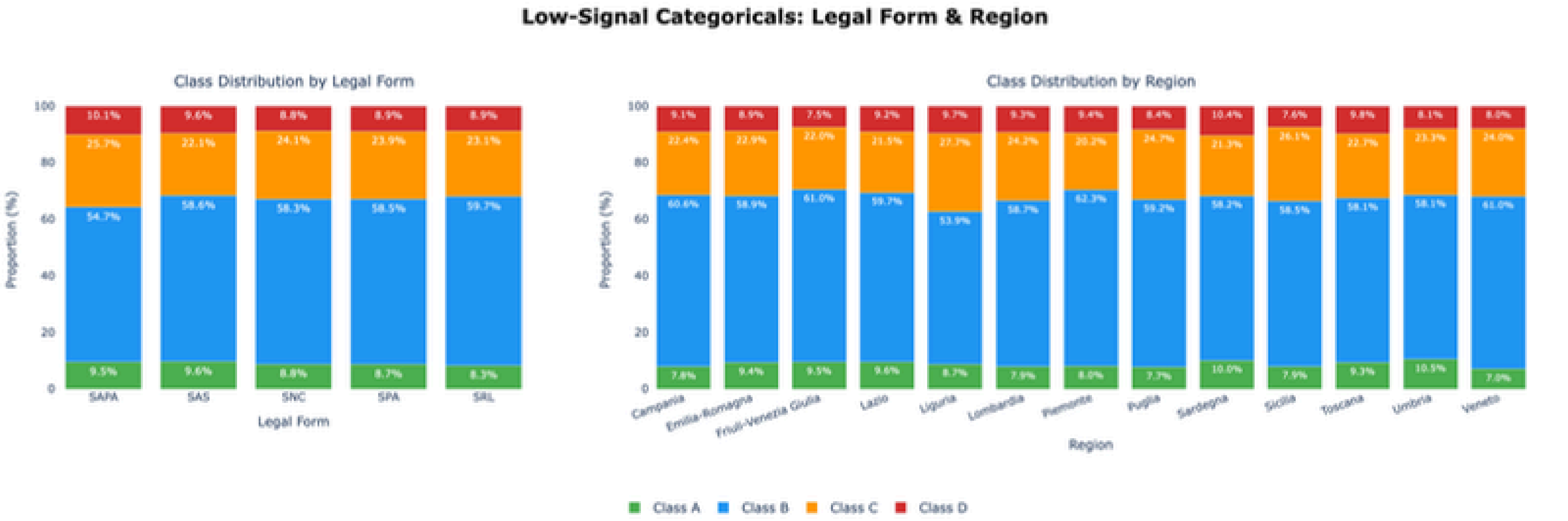
TEMPORAL STABILITY (2018–2021)



No **COVID-19 spike** detected

CATEGORICAL FEATURES: LOW STANDALONE SIGNAL & STRATEGY



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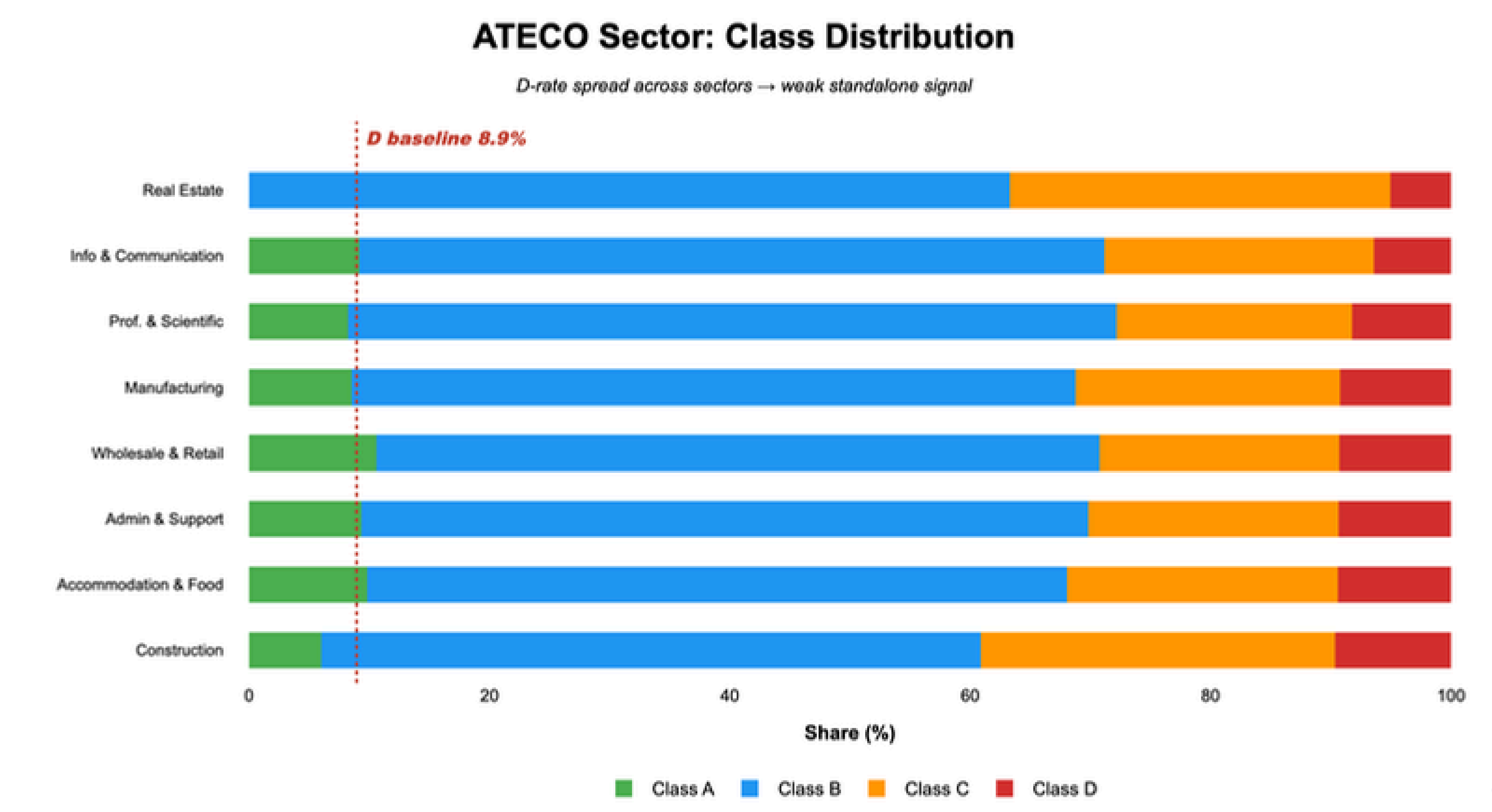

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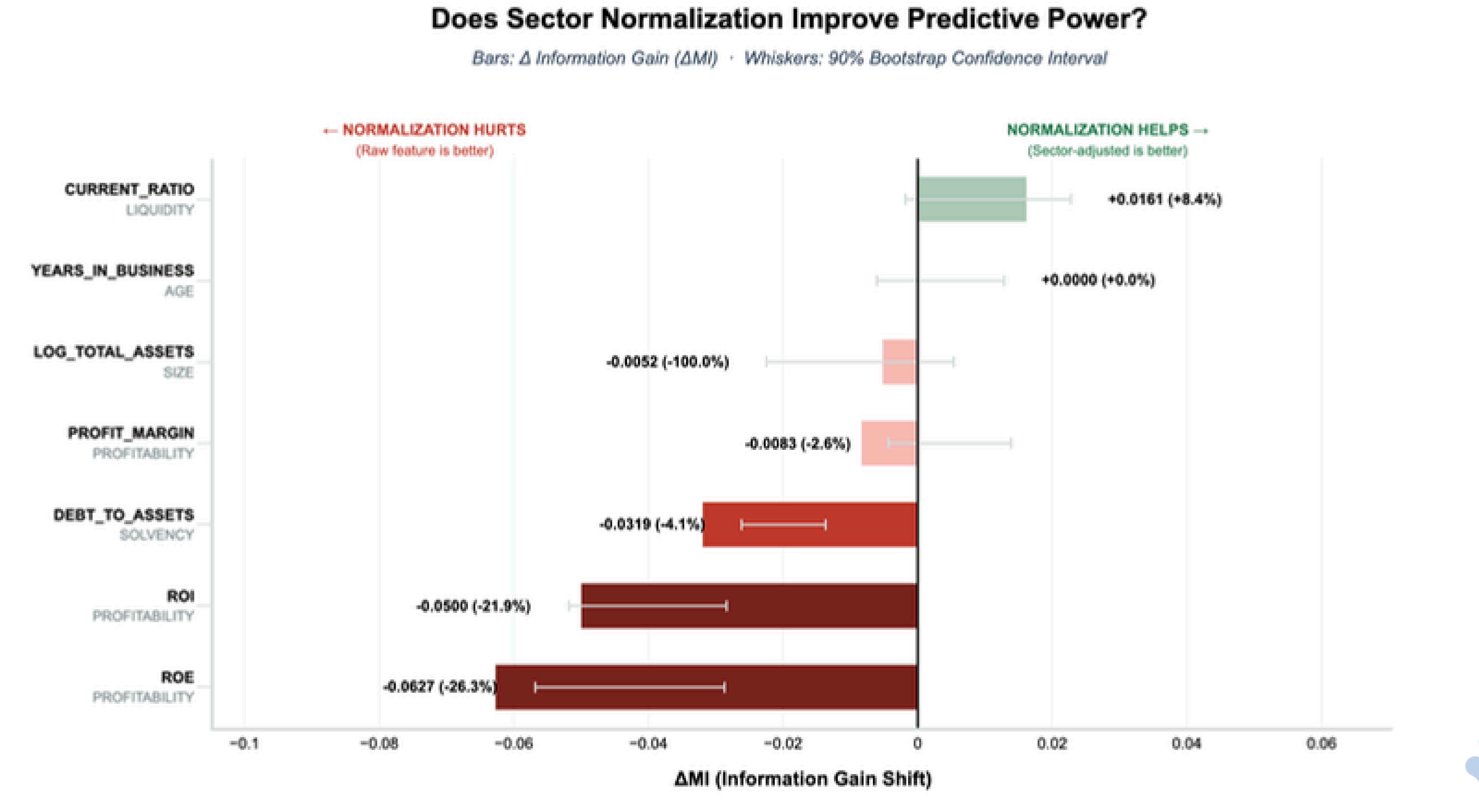
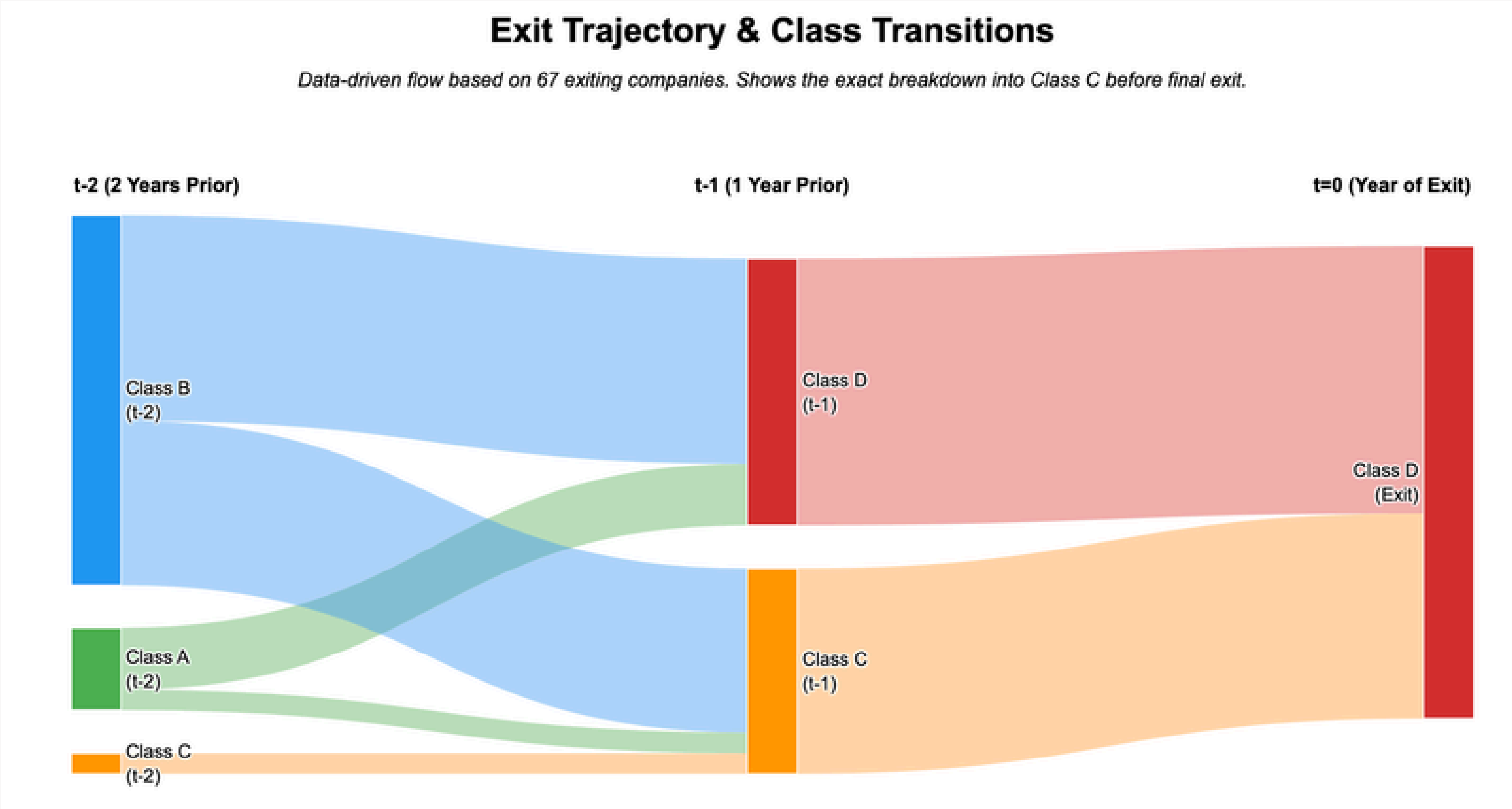


THE BIAS IS THE FEATURE



Credit Migration Matrix: Year-over-Year Transition Probabilities

n=8,896 transitions · Rows sum to 100% · High-Contrast Intensity Applied

Year t	→ A	→ B	→ C	→ D	→ EXIT
From A	11.0%	60.6%	20.8%	7.6%	—
From B	8.2%	58.4%	24.6%	8.8%	—
From C	8.2%	58.2%	24.2%	9.3%	—
From D	7.6%	54.4%	18.5%	11.1%	8.5%

■ Financial Upgrade ■ Credit Stability ■ Financial Downgrade ■ Default / Exit

- 100% of exits are preceded by a Class D state.
- Distress is a traceable process, not a random event.

- Mean Reversion Trap: ~**60% flow to Class B** ~9% drop to Class D
- **8.5% Exit (Row D only)**

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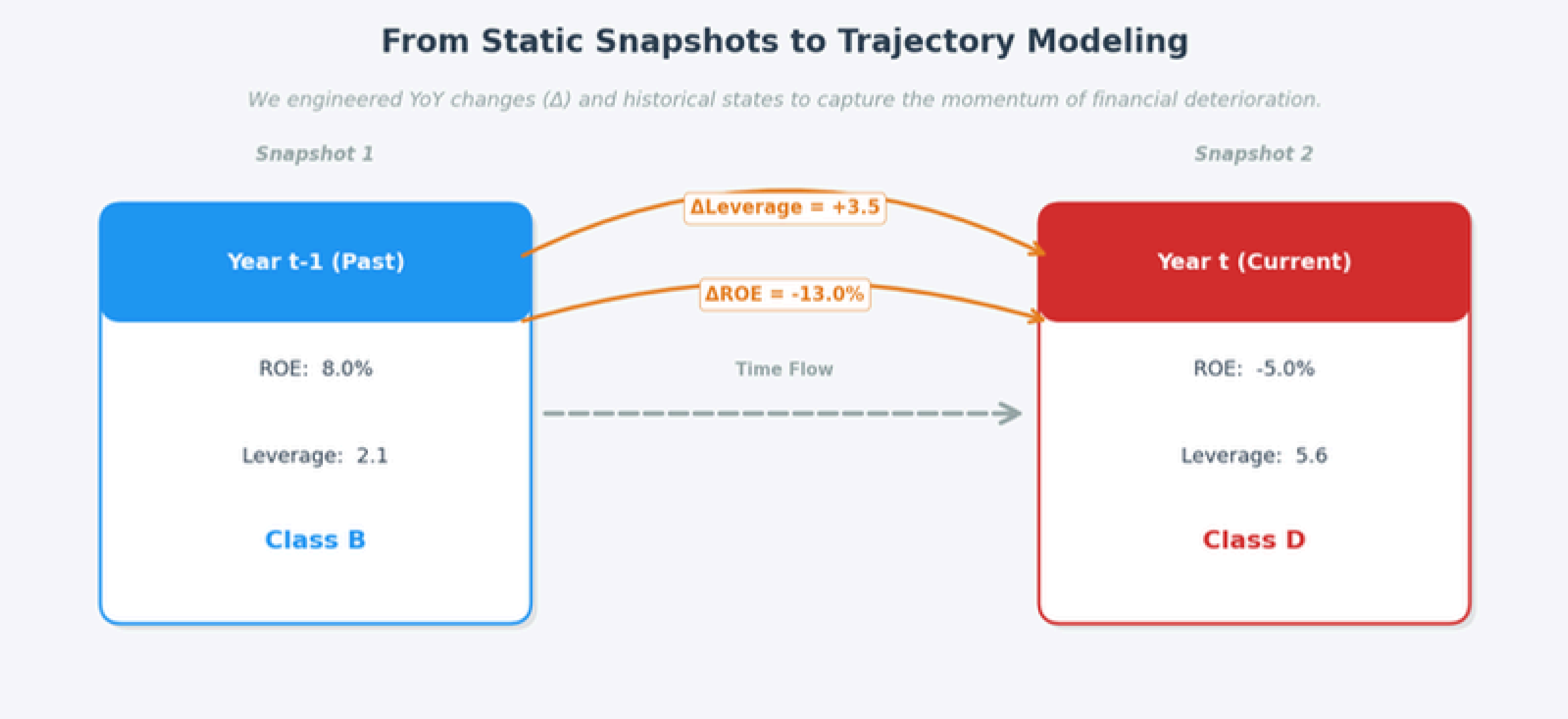

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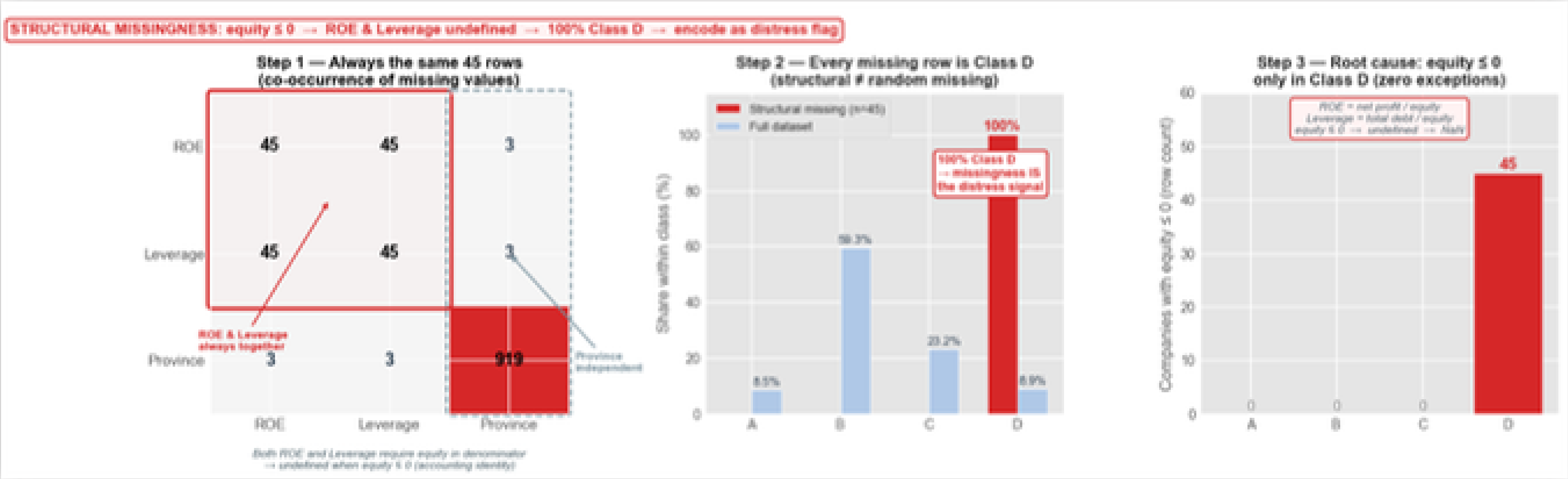
MISSING VALUES:
TWO MECHANISMS, TWO RESPONSES



These are **not ordinary missing values:** they encode structural financial distress and should be preserved as a **predictive signal**

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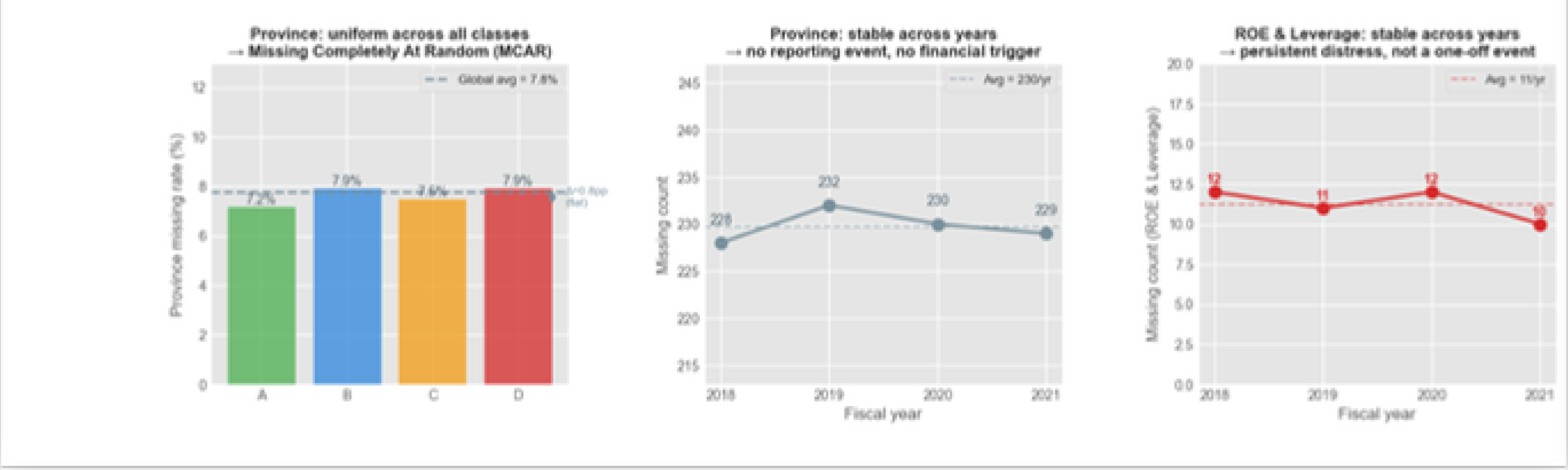
MISSING VALUES: TWO MECHANISMS, TWO RESPONSES



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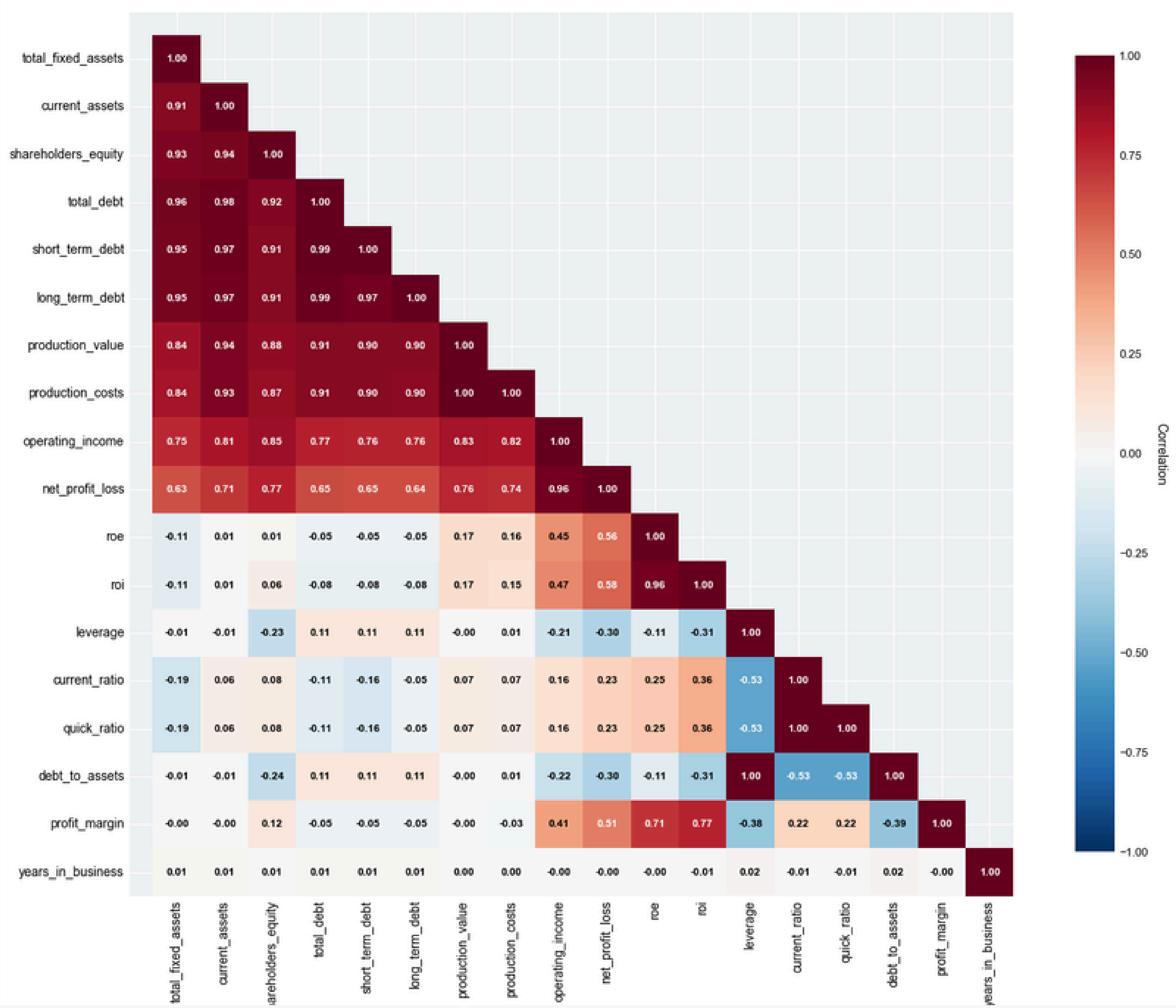
FEATURE CORRELATION MATRIX



Key Findings

- Size variables & 3 ratio pairs redundant → **dropped**
- ROE, ROI, profit margin independent → **retained as core features**
- Eliminates mathematical identities (DTA/Leverage) to prevent coefficient inflation, ensuring a parsimonious, audit-ready model

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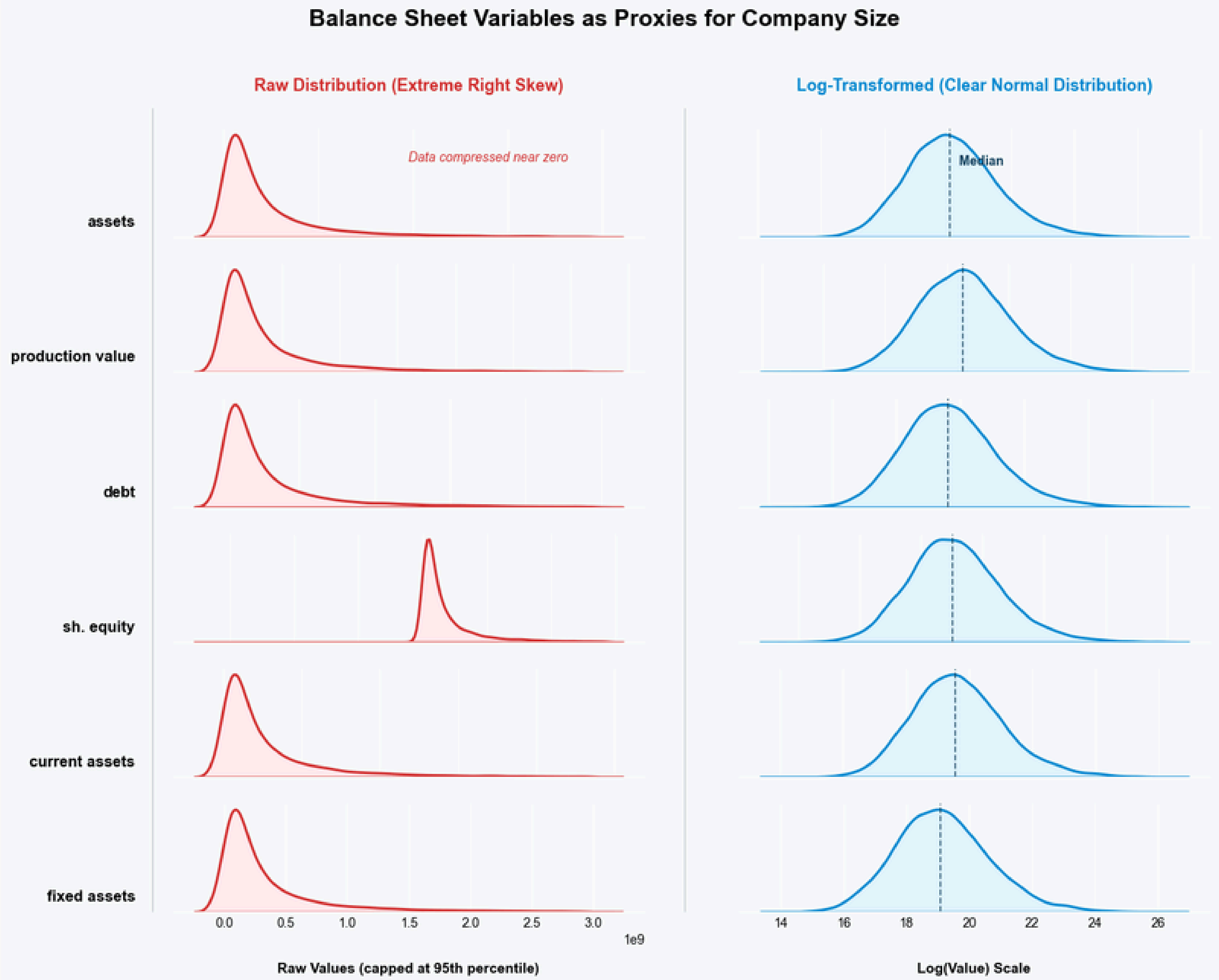
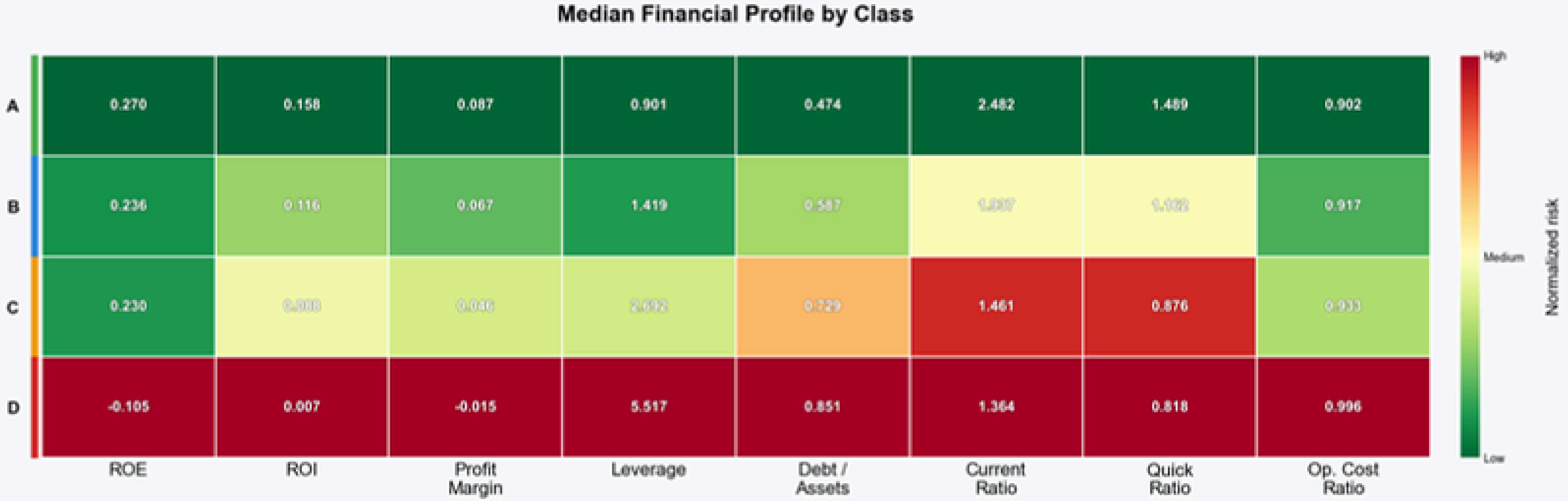
DISTRIBUTION ALIGNMENT & FEATURE STABILITY



Key Findings

- Raw monetary variables are dropped; only a single log-size proxy is kept

MEDIAN FINANCIAL PROFILE BY HEALTH CLASS



- Ratios worsen consistently **from A → D**, showing a clear **risk gradient**
- Profitability collapses and leverage spike in class D
- Confirms A–D is ordered and ratios matter more than raw size

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THANK YOU

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